In class lab #8 for Numerical Analysis

Your name: _____

scan and submit before 23:00

Part I: Newton's Divided Differences (10 points)

Problem 1. Step 1. Click on New to create a new document and type the code below. Once you are done, save it by the name `newtdd.m`:

```
%Program 3.1 Newton Divided Difference Interpolation Method
%Computes coefficients of interpolating polynomial
%Input: x and y are vectors containing the x and y coordinates
% of the n data points
%Output: coefficients c of interpolating polynomial in nested form
%Use with nest.m to evaluate interpolating polynomial
function c=newtdd(x,y,n)
for j=1:n
v(j,1)=y(j); % Fill in y column of Newton triangle
end
for i=2:n % For column i,
for j=1:n+1-i % fill in column from top to bottom
v(j,i)=(v(j+1,i-1)-v(j,i-1))/(x(j+i-1)-x(j));
end
end
for i=1:n
c(i)=v(1,i); % Read along top of triangle
end % for output coefficients
```

Step 2. Click on New to create a new document and type the code below. Once you are done, save it by the name `nest.m`:

```
%Program 0.1 Nested multiplication
%Evaluates polynomial from nested form using Horner's Method
%Input: degree d of polynomial,
% array of d+1 coefficients c (constant term first),
% x-coordinate x at which to evaluate, and
% array of d base points b, if needed
%Output: value y of polynomial at x
function y=nest(d,c,x,b)
if nargin<4, b=zeros(d,1); end
y=c(d+1);
for i=d:-1:1
y = y.*(x-b(i))+c(i);
end
```

Step 3. (10 points) Type

```
>> x0=[0 2 3];
>> y0=[1 2 4];
>> c=newtdd(x0,y0,3);
>> x=0:.01:4;
>> y=nest(2,c,x,x0);
>> plot(x0,y0,'o',x,y)
```

Part II: Numerical Differentiation (80 points)

problem 1. (20 points)

Develop a second-order method for approximating $f^{'}(x)$ that uses the data $f(x-h)$,$f(x)$, and $f(x+3h)$ only. (Hint: the method of undetermined coefficients)

problem 2. (10 points)

Find the error term and order for the approximation formula

$$f^{'}(x) = \frac{4f(x+h) - 3f(x) - f(x-2h)}{6h}$$

problem 3. (30 points)

Make a table of the error of the three-point center-difference formula for $f^{'}(0)$,where $f(x) = sinx - cosx$,with $h = 10^{-1}, \cdots, 10^{-12}$,and the table includes h, D_h^C, $Error$, $Ratio$. Then draw a plot of the error results. Does the minimum error correspond to the theoretical expectation?

problem 4. (30 points)

Consider the second order linear two point boundary value problem,

$$\begin{cases} -u^{''}(x) + 2u(x) = 1, x \in [0,1] \\ u(0) = 0, u(1) = 1 \end{cases}$$

give the numerical differectiation formulas and solve it.